

CRYSTAL NG SHU LU

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TECHNICAL SKILLS

Languages: C++, Java, Python, JavaScript, TypeScript, Haskell, HTML, GDScript

Developer Tools: GitHub, Flask, JQuery, SQL, AJAX, CSS Tailwind, Godot, Unreal Engine

Libraries: MySQL, TypeScript, Pygame, NumPy, Matplotlib, SQLite

EDUCATION

University of Warwick

BSc Computer Science - Achieved a First in the 1st Year

Sep. 2024 – Jun. 2027

Highgate School

A-Levels: Computer Science (A*), Maths (A*), Further Maths (A), Physics (A). GCSE: Eight 9's, two 8's

Sep. 2009 – Jul. 2024

EXPERIENCE

Software Engineer

University of Zurich

Aug. 2025 – Sept. 2025

Remote

- Contributed to 'Clear Skies,' an open-source app; honed an understanding of low-level programming principles by programming without the aid of a pre-built game engine (using Python, PyGame).
- Worked on over **20** bug-fixes and QA implementations at the end of the projects development to ensure quality for the **600+** people playing the game during a psychology study.
- Worked with an international team of **15+** people to deliver the project in time for tight deadlines of when the study would occur.

Software Engineer (Paid Internship)

Warwick Manufacturing Group, WMG

Jun. 2025 – Aug. 2025

Coventry, United Kingdom

- Developed an autonomous car driving simulator where users must navigate emergency vehicles in user-created scenarios.
- Designed and implemented an algorithm for rule-based vehicle placement, ensuring emergency vehicles were strategically placed based on spatial logic such as manhattan distance and ability to reach a randomly generated road accident.
- Created and programmed an interactive tutorial to teach users how to use the simulation by guiding them through creating a scenario; the easy to apprehend UI increased user understanding of the simulation by **85%**.
- Created comprehensive **50** page documentation for the entire projects codebase, outlining the function of key files, systems, and classes in the code with tables, diagrams, and explanations.

Game Developer and Founder

Prelude Game Studios

Dec. 2024 – Present

London, United Kingdom

- Founded and managed an independent game studio focused on innovative 2D and 3D game development.
- Sole developer of debut title "Amaz'd," engineered using OOP principles using Python and Pygame, achieving positive global reviews and over **200** players on Steam.
- Designed advanced features such as custom enemy AI and procedural level generation via depth-first maze creation for gameplay depth.
- Built and maintained the entire development pipeline for all studio projects, including testing in pytest, debugging, and deployment for cross-platform release by using the steam SDK.

Graphic Designer and Video Editor

Little Big Bell

Apr. 2021 – Apr. 2022

London, United Kingdom

- Produced and edited high-quality interior design and lifestyle video content, including branded segments for clients such as Amazon.
- Delivered engaging, public-facing videos through editing and design on Photoshop and Vegas Movie Studio, accumulating **20,000+** Instagram views per video.

PROJECTS

Poker Hand Evaluation & AI Engine | Haskell, Functional Programming

Dec. 2024 – Feb. 2025

- Developed the back-end logic for a poker-like game featuring a comprehensive hand evaluation and scoring system, by using functional programming concepts such as higher-order functions for code abstraction and reusability.
- Developed a custom AI opponent capable of consistently outperforming baseline strategies, achieving **80%+** optimal play by simulating outcomes and leveraging probabilistic heuristics such as flush fallacy detection.
- Applied game-theoretic decision-making principles to construct a codebase transferable to fintech contexts such as investment portfolio optimisation or algorithmic trading strategy evaluation.

Banking Financial Literacy App - <WHACK> Hackathon Title Winning App | Python, Pygame, Github

Oct. 2024

- Led a team of four in designing and developing an educational app to teach financial literacy to children, winning out of **150** contestants.
- Built a virtual stock market simulator using Pandas and a Python-based savings account model.
- Optimised team performance by assigning roles to leverage individual strengths and maximise productivity under tight deadlines.
- Pitched the project to Capital One representatives in a time-pressured presentation, demonstrating clear and efficient communication.

NVIDIA Jetson AI Certification | Linux, Python, Neural Networks

Jun. 2023 – Jul. 2023

- Developed and trained an AI detection model utilising the ResNet18 neural network architecture on a Jetson Nano, enabling colourblind individuals to differentiate Skittle flavours based on colour recognition.
- Gained hands-on experience in machine learning by optimising model accuracy through iteratively training epochs **42** times and evaluating performance, with the model giving **96%** accuracy for recognising the correct skittle colour.

Other Projects

- Robot Maze Solving Algorithm** (Java, A* Algorithm) | **Low-Latency Movie Data Indexing App** (Java, Algorithmic Optimisation) | **3D Aim Training Simulation** (C++, Unreal Engine) | **Grid-Based Game Parser** (Haskell, Parsing) | **Pixel Art Gallery Social Media Site** (Flask, AJAX, JQuery, TypeScript, MySQL) | **Binary-Star Exoplanet Detection & Classification Model** (Python, Matplotlib)